

State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts

JetBrains Survey of 23,262 Developers Worldwide

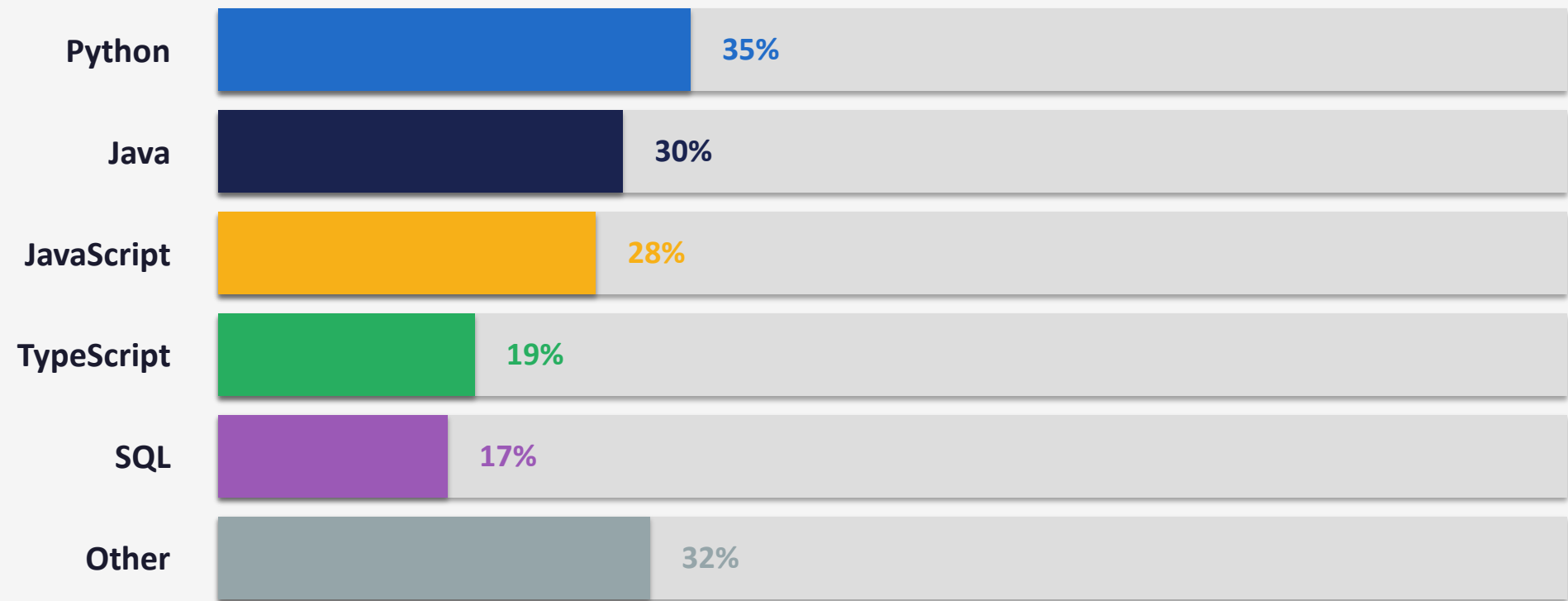
For Engineering Leaders, CTOs & Developer Experience Teams

Python Closes the Gap: 8 Years of Language Usage Shifts

Language	2017	2020	2022	2024	Trend
JavaScript	65	70	65	61	Declining from peak
Python	32	55	53	57	Steady growth
HTML/CSS	60	61	54	51	Gradual decline
SQL	42	56	49	48	Plateaued
Java	47	54	48	46	Slow decline
TypeScript	12	28	34	37	Strong growth
Go	8	19	19	18	Plateaued
Rust	0	7	9	11	Steady growth
PHP	30	27	20	17	Significant decline
Kotlin	2	17	16	14	Plateaued

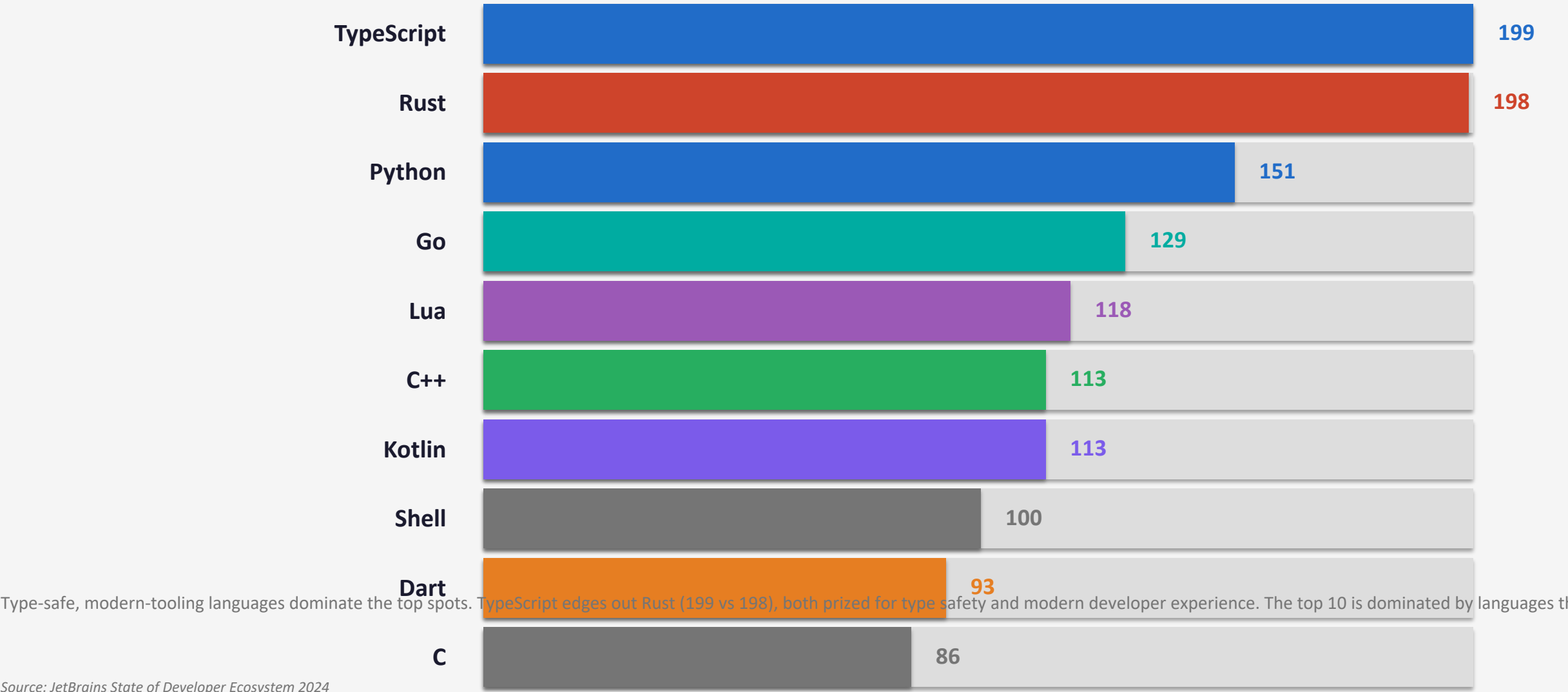
Source: JetBrains State of Developer Ecosystem 2024 | Base: 23,262 developers worldwide

Python Is Now the #1 Primary Language at 35%



Python leads as primary language for 35% of developers. Despite JavaScript being most used overall (61%), Python's dominance in AI/ML and data science has made it the most common primary language. A

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



Type-safe, modern-tooling languages dominate the top spots. TypeScript edges out Rust (199 vs 198), both prized for type safety and modern developer experience. The top 10 is dominated by languages that

Source: JetBrains State of Developer Ecosystem 2024

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Application Domain	Leading Language	Share
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs	Python	38%

Language choice is driven by domain, not general preference. Python dominates AI/ML and data processing; TypeScript/JavaScript own UI; Java leads server-side APIs and workflows; C++ leads systems programming.

What This Means for Your Technology Strategy in 2025

1

Python for AI/ML Teams

Python is the undisputed leader in AI/ML, data science, and data processing. Invest in Python expertise to stay competitive in the AI era.

2

TypeScript Is the Future of Frontend

With 62% dominance in UI development and the highest developer satisfaction score (199), TypeScript is the strategic choice for frontend and full-stack investment.

3

Rust & Go Are the Growth Plays

Rust (198 satisfaction) and Go are climbing steadily and are the right targets for infrastructure, cloud-native, and systems programming roles.

4

PHP & Ruby Hiring Pools Shrinking

PHP dropped from 30% to 17% (2017–2024). Ruby fell from 10% to 4%. Both languages are in long-term decline — factor this into hiring and legacy modernization plans.

5

Satisfaction Correlates with Type Safety

TypeScript, Rust, and Kotlin top satisfaction scores. Developer experience and type safety drive retention. Factor language satisfaction into your stack decisions.